

Fig. 3: Captured scene rendered from reconstructed 3D geometry mimicking the surgeon (Source: <http://graphics.cs.berkeley.edu/research/tele/teleImmersionApplications.html>)

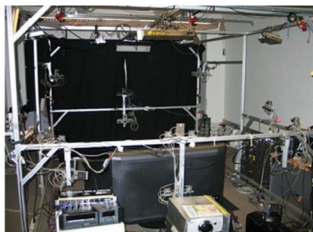


Fig. 4: The Tele-immersion Lab at University of California, Berkeley, the USA (Source: <http://tele-immersion.citris-uc.org/lab>)

jections as if these were real. And, desktop supercomputers would complete trillions of calculations needed to create a holographic environment. Another possibility to support these environments would be a network of computers that split the total power.

Application areas

Tele-immersion can benefit a host of areas from being able to meet relatives you are unable to visit to treating very remotely-located critical patients. Here are the main application areas:

In medicine. Tele-immersion can prove to be tremendously helpful

and even life-saving in the area of medicine. Traditionally, medicine has been taught and practiced face to face. However, with tele-immersion, 3D surgical learning for virtual operations is now possible. In times to come, we will be able to tele-immersion a surgeon into an operation theatre to carry out a surgery (Fig. 3).

This could imply that many lives could be saved, especially if there are patients who are in dire need of a certain process or equipment that is not available at their location. Some loose ends that need to be tied are the legal implications in case of any system or tele-immersed surgeon failure. Also, since the doctor carrying out the operation is actually not physically present, how effective will he or

she be? Will the doctor be able to gauge what is transpiring in the back-end of medical services? Will the patient feel comfortable going through a critical procedure in this manner?

High-resolution 3D patient images have now become quite common, and it is vital to make this patient-specific data quickly available and usable to many specialists at various geographical locations. While web based systems can provide volume or surface rendering of medical data over networks with low fidelity, these are not able to manage stereoscopic visualisation or huge datasets effectively. This is where

state-of-the-art tele-immersion techniques and high-speed networks are required to produce an environment for geographically-distributed clinicians to share these huge datasets in real-time basis.

In education. Tele-immersion allows geographically-distributed students to come together in a single environment. This is particularly important given the lack of specialist faculty at all required sites, and hence the need to resort to distance education. This technology will result in a much higher level of collaboration among educational institutes. Moreover, it will also help in mitigating the risk, for example, budding surgeons will get a much better experience and training w.r.t. complex situations before actually operating on their first patients. In the similar manner, a pilot will experience an almost life-like flight simulation before his or her maiden flight. School students will be able to access data or govern a telescope from a remote location.

Research activities at the Tele-immersion Lab at University of California, Berkeley, the USA (Fig. 4), include human-robot co-operation, human activity recognition from multi-modal data, development of individualised musculo-skeletal models, quantification of human performance, remote monitoring in health care and its privacy and security considerations, and modelling of driver interaction in semi-autonomous vehicles. The lab supports research in 3D stereo reconstruction, tele-immersion, motion capturing, remote monitoring and human-movement analysis.

The tele-immersion apparatus with 48 cameras has been developed to capture 3D data of users with accuracy of about 1cm to 2cm. Cameras are linked to 12 servers that process the stereo reconstruction algorithm, allowing full-body 3D reconstruction of user(s). Some applications that the lab is presently concentrating on include collaborative work on 3D data sets, such as medical (MRI, 3D